Data Analyst Professional Findings Report

- **Project Title:** Analysis of Technology and Developer Trends (Stack Overflow 2023)
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- **Course:** IBM Data Analytics Professional Certificate (Coursera)
- **Date:** March 15, 2026

Slide 1: Title Slide

- **Project Name:** Global Developer Survey Insights .2023
- **Student Name:**.firahS-IA namharludbA demmahoM halludbA
- **Date:** March .2026 ,15
- **Affiliation:**.noitazilaicepS scitylanA ataD MBI

Slide 2: Outline

- **Section 1 :**Executive Summary & Introduction.
- **Section 2 :**Methodology & Data Wrangling.
- **Section 3 :**Technical Findings (Languages, Databases, Job Roles)
- **Section 4 :**Advanced Insights (Compensation, AI Tools, Remote Work.)
- **Section 5 :**Looker Studio Dashboards.
- **Section 6 :**Overall Findings, Conclusion, and Appendix.

Slide 3: Executive Summary

- **Dominance:** JavaScript and SQL remain the bedrock of the industry for current deployment.
- **Future Shift:** ekil segaugnal ecnamrofrep-hgih drawot devresbo si noitargim tnacifingis A .oG dna tsuR
- **Database Leadership:** PostgreSQL has emerged as the most desired database for 2024 and beyond.
- **AI Integration:**45% of developers are already using AI tools to augment their coding productivity.
- **Demographics:** The market is driven by young professionals (34 -25 years old)with a strong preference for Hybrid work.

Slide 4: Introduction

- **Objective:** To transform raw survey data into actionable insights for tech stakeholders.
- **Target Audience:** CTOs, Hiring Managers, and Aspiring Developers.
- **Value Proposition:** This report identifies the gap between current skills and future market demands, enabling better career and project planning.

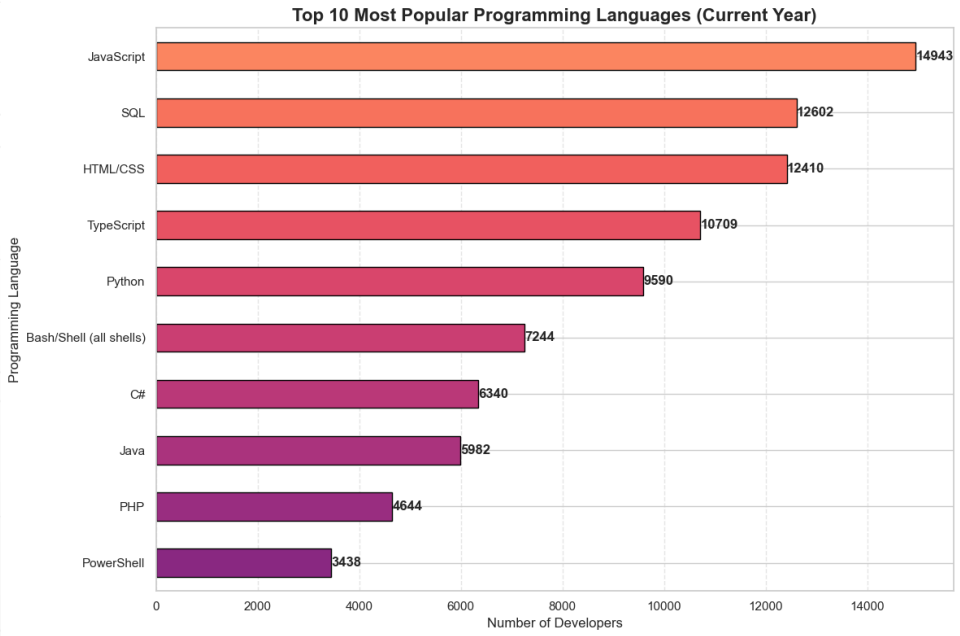
Slide 5: Methodology

- **Data Source:** Stack Overflow Annual Developer Survey (+90,000)2023 .tnednopser

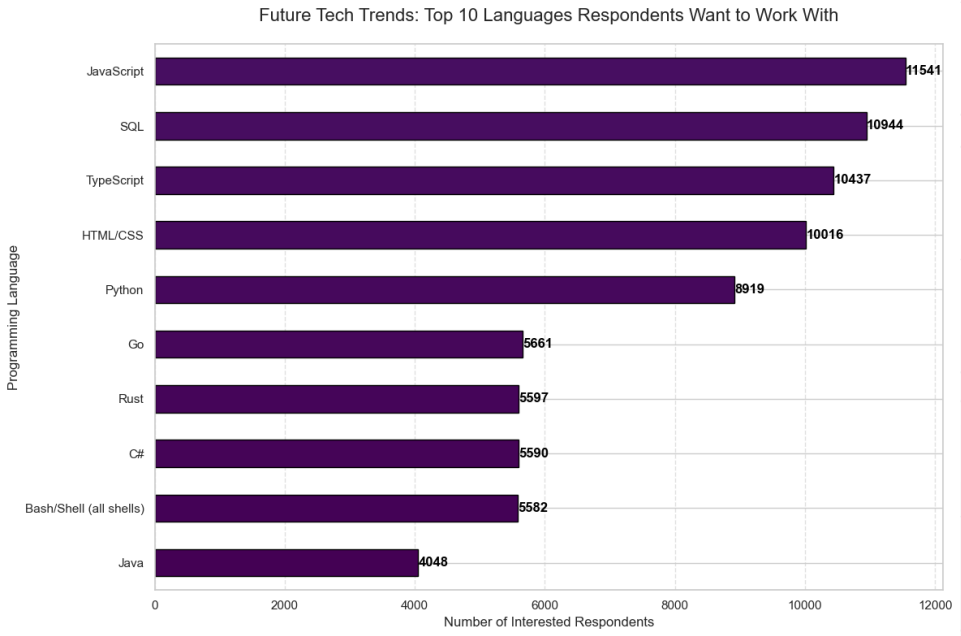
- **Processing Steps:**
  - **Data Cleaning:** Filtered for the Top 10 results in each category to eliminate noise.
  - **Imputation:** Missing values were filled using the **Mode** to maintain statistical integrity.
  - **Cross-Referencing:** Validated findings against the job-postings.xlsx file to ensure the supply of skills matches market demand.

Slide 6 & 7: Results - Programming Language Trends

- Top 10 Most Popular Programming Languages



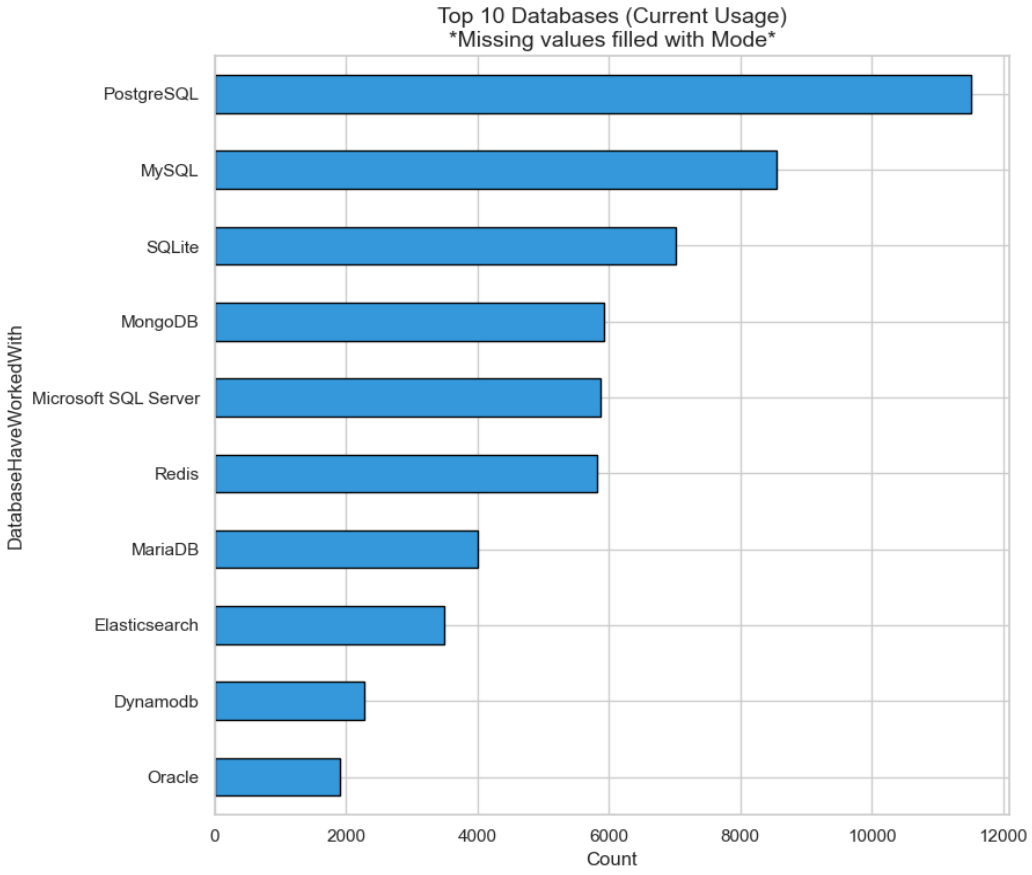
- Future Tech Trends Top 10 Languages Respondents Want to Work With.png

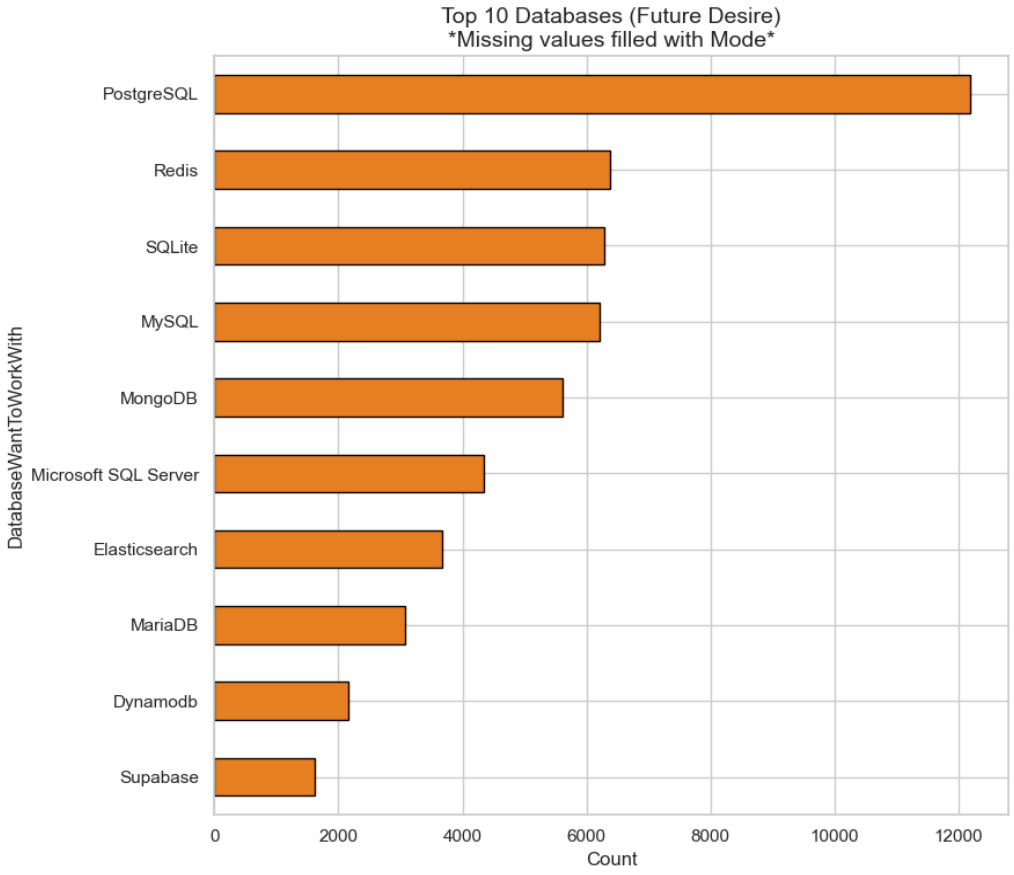


Slide 6 & 7: Results - Programming Language Trends

- **Detailed Analysis:**
  - **Current Usage:** JavaScript leads with 14,943 users, followed by SQL and HTML/CSS, confirming the continued importance of Web Development.
  - **Future Desire:** While JavaScript stays high, **Rust** and **Go** show a massive surge in interest. This indicates a shift toward "Safe" and "Performant" system-level programming.
- **Implication:** Developers should consider learning **TypeScript** and **Python** to remain competitive in the coming year.

Slide 8 & 9: Results - Database Trends

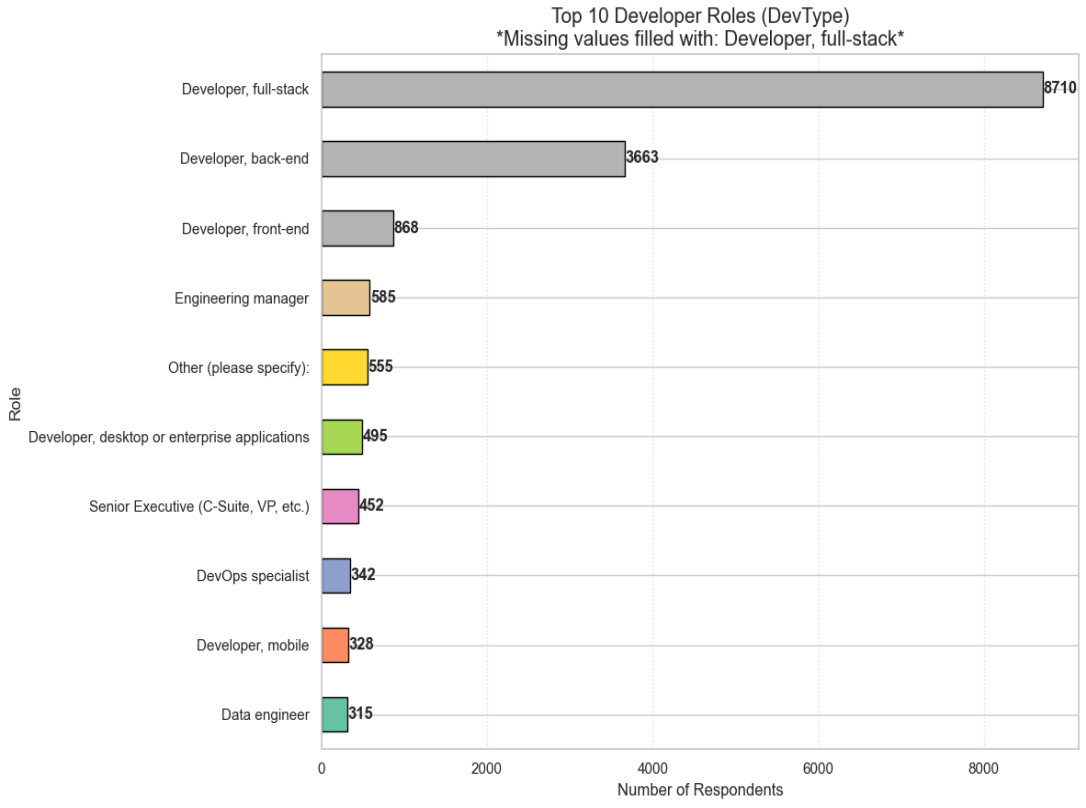




Slide 8 & 9: Results - Database Trends

- **Detailed Analysis:**
  - **PostgreSQL** is the clear winner, being both the most used and the most desired (stnednopser (+12,000.
  - Traditional systems like Microsoft SQL Server are seeing a decline compared to open-source flexible solutions like **Redis** and **Supabase.**
- **Implication:** The industry is moving toward open-source ecosystems that offer better scalability and lower licensing costs.

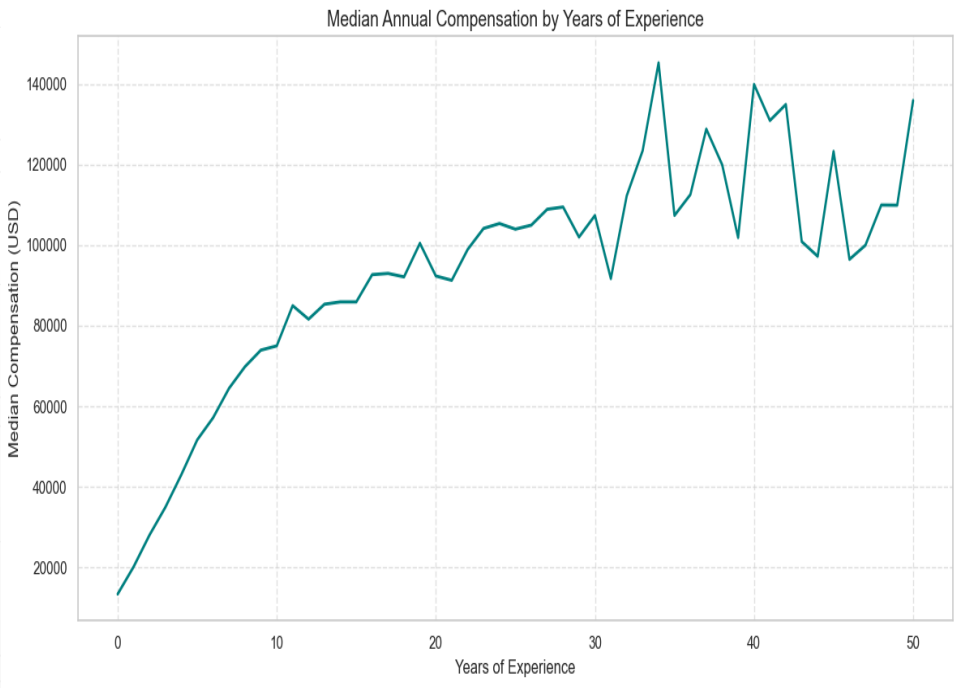
Slide 10: Job Postings Visualization



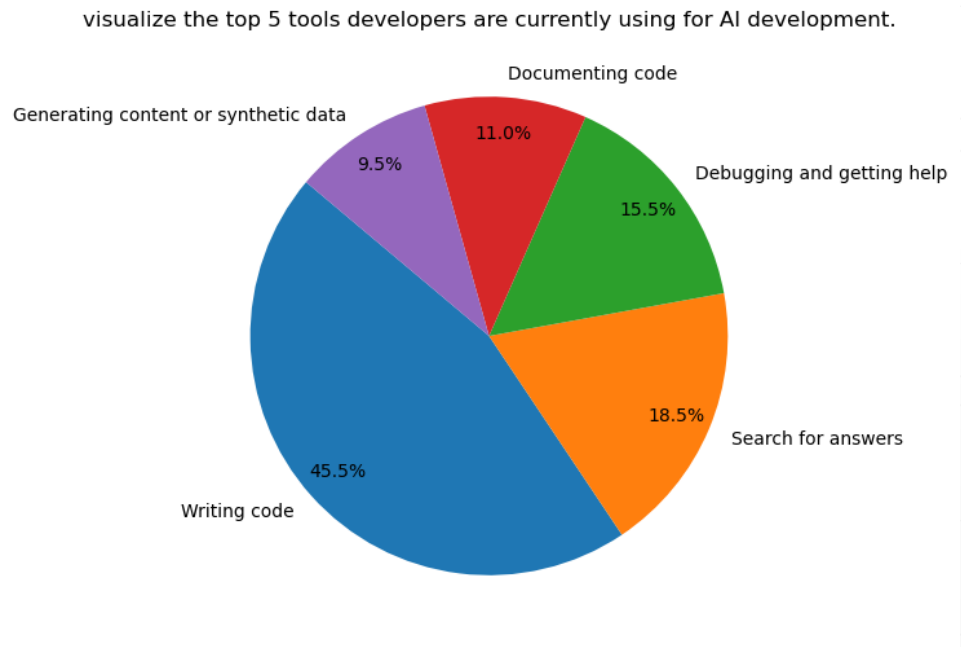
- **Analysis:** Derived from job-postings.xlsx .(gnidnecseD detroS)
- **Insight: Full-stack Developer** roles are the most frequent, which perfectly aligns with the high usage of JavaScript/SQL. **Data Engineering** follows closely, highlighting the need for professionals who can build robust data pipelines.

Slide 11: Innovative Insights - Compensation, AI, and Industry

- Median Annual Compensation by Years of Experience



- visualize the top 5 tools developers are currently using for AI development



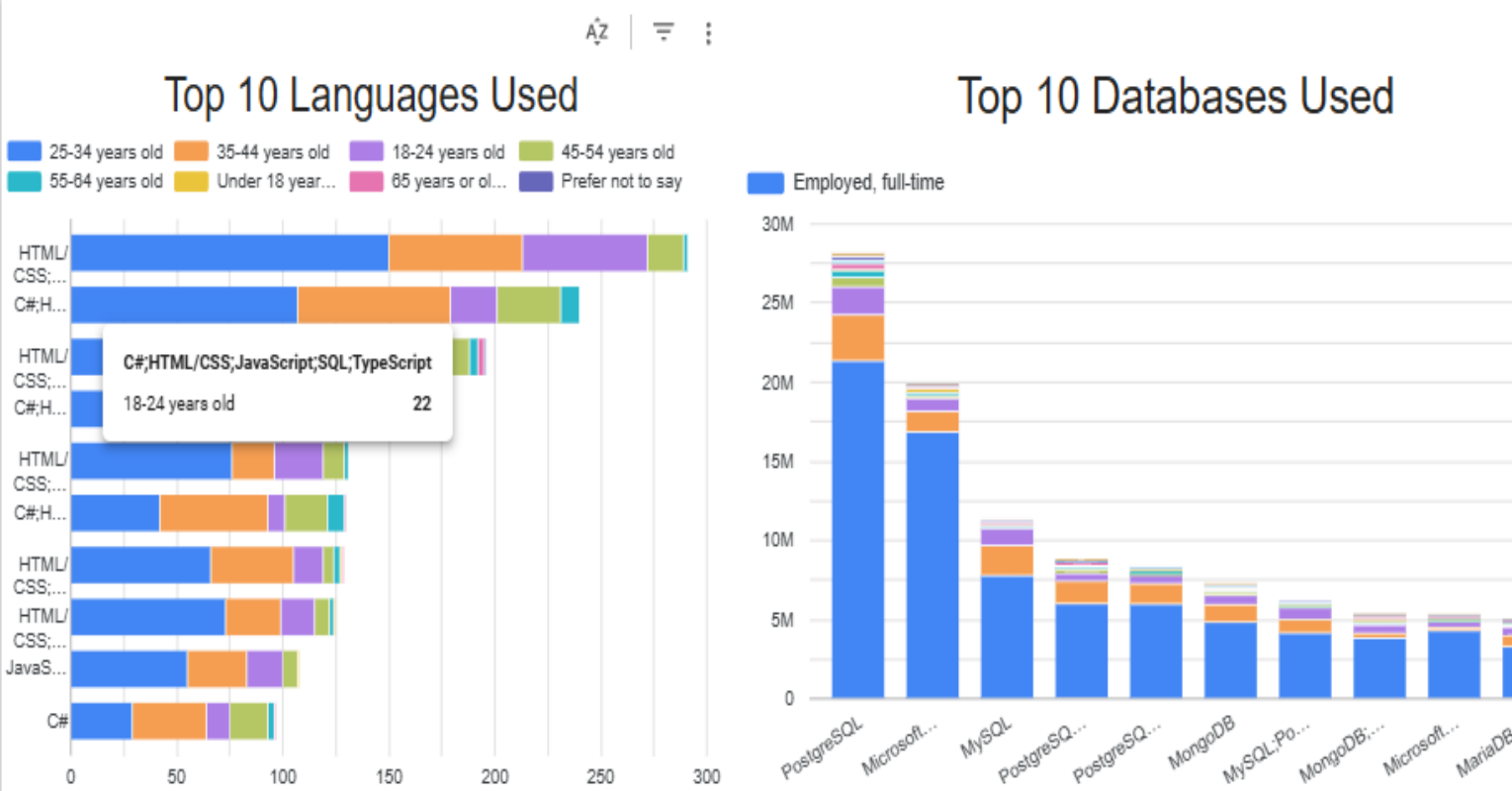
Slide 11: Innovative Insights - Compensation, AI, and Industry

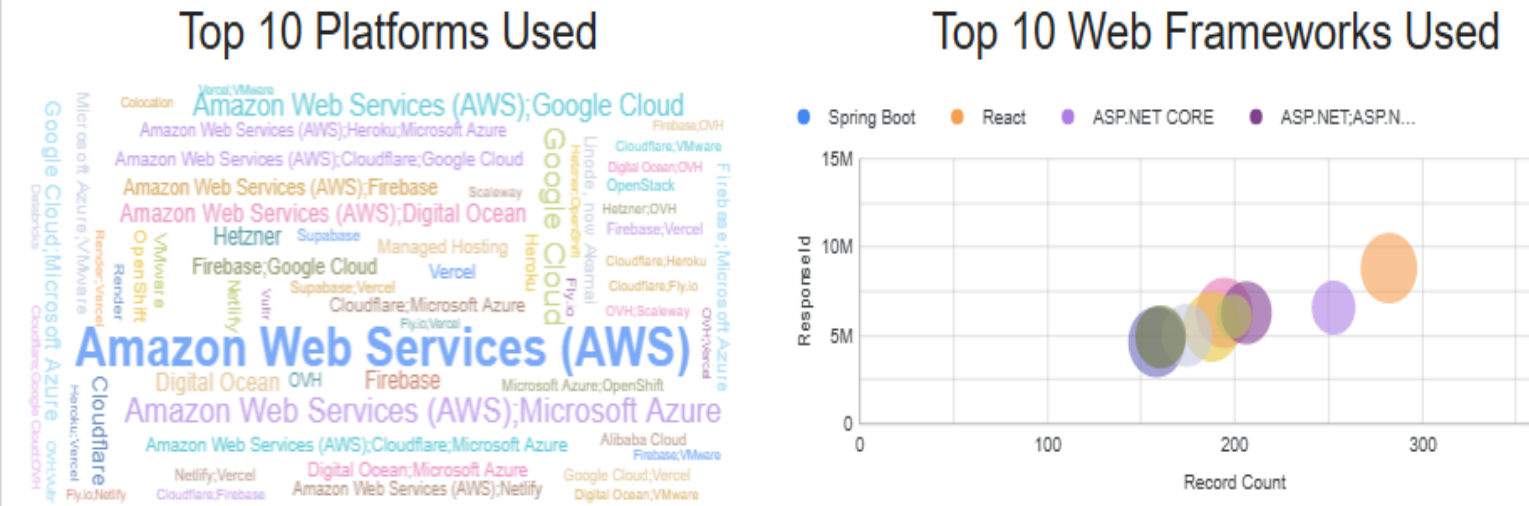
- **Analysis:** There is a direct correlation between **Experience** and **Compensation**10 tsrif eht gnirud yralas ni esaercni peets a htiw ,years.
- **AI Integration:**45.5% of AI usage is for **Writing Code** yranoitulover a si sihT . evitcudorp erom eb ot ylekil era gnidoc detsissa-IA retsam ohw srepoleved ;tfihs .seiralas rehgih dnammoc dna

Slide 12, 13, 14: Looker Studio Dashboards

Slide 12

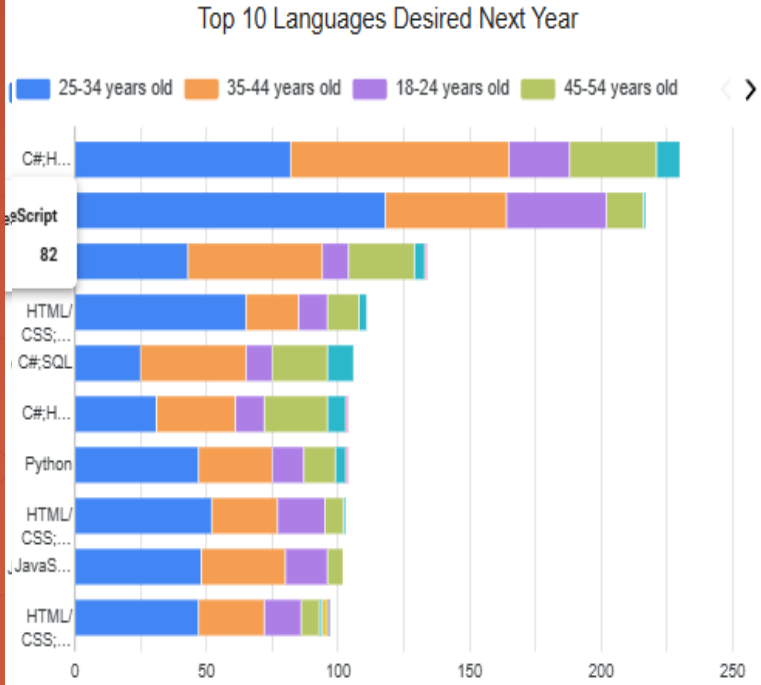
Shows AWS and React dominance

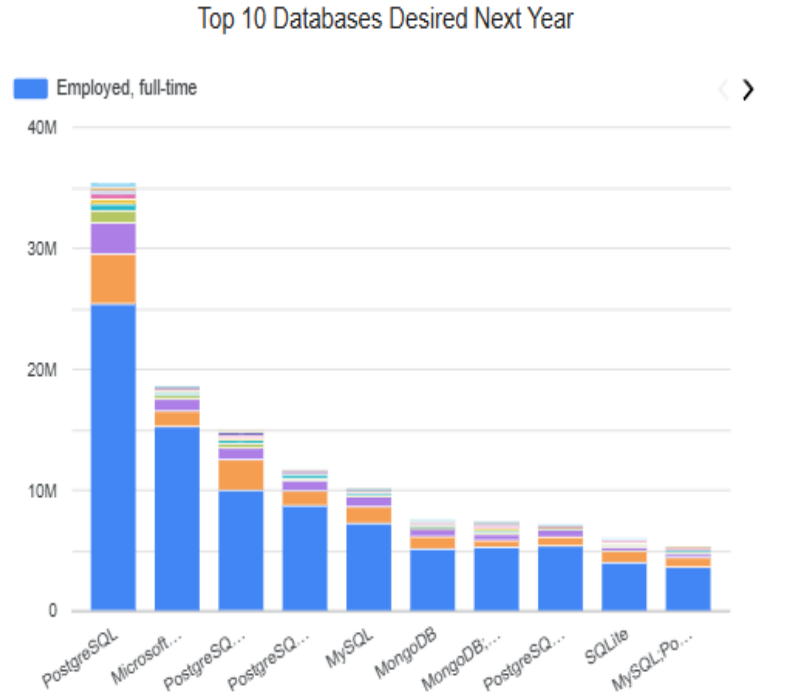


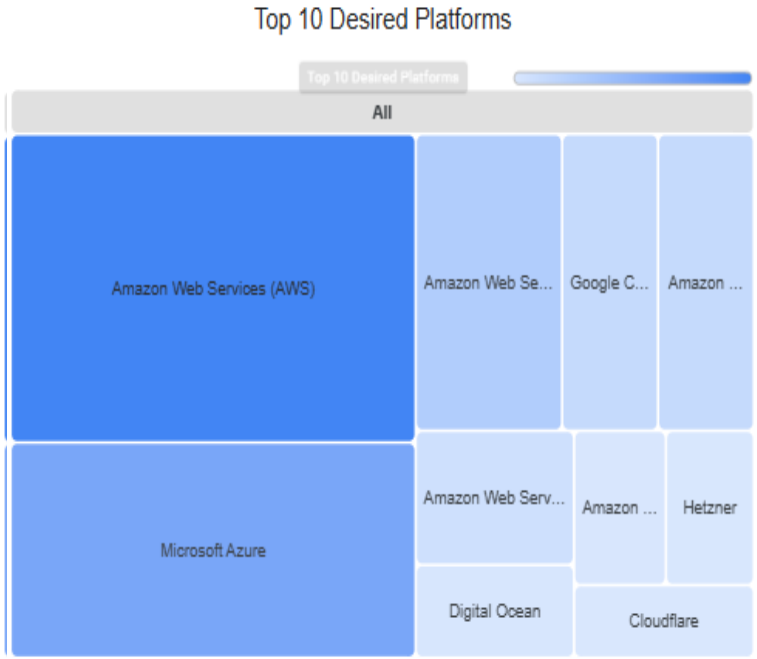


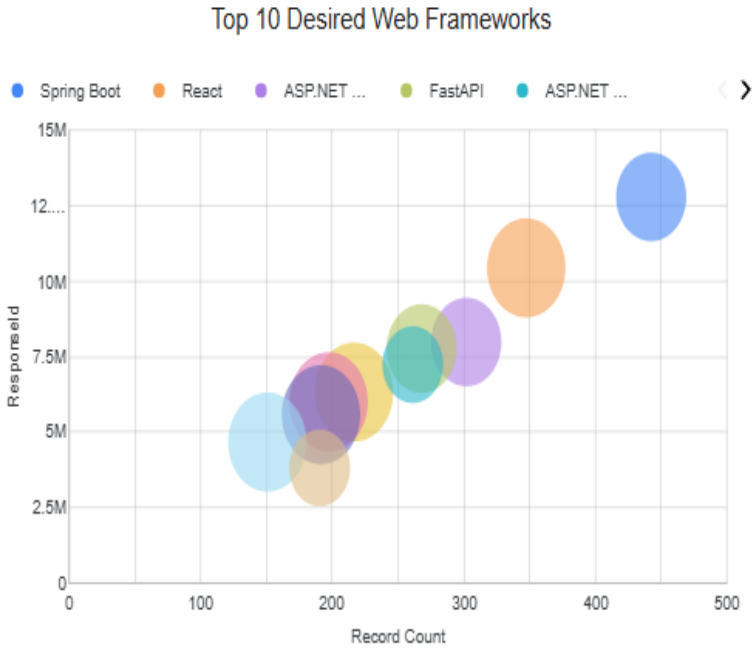
Slide 13

Focuses on Cloud-native tools and Next.js



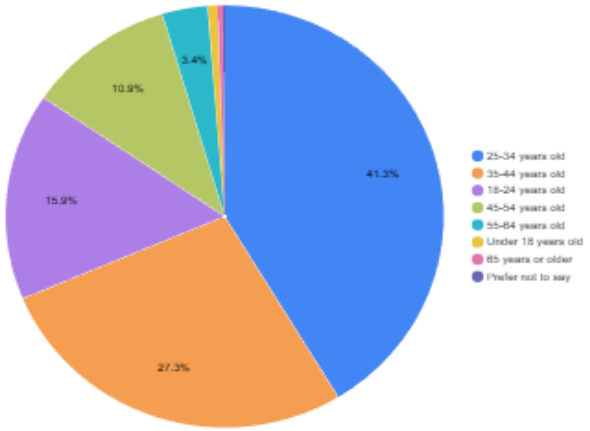


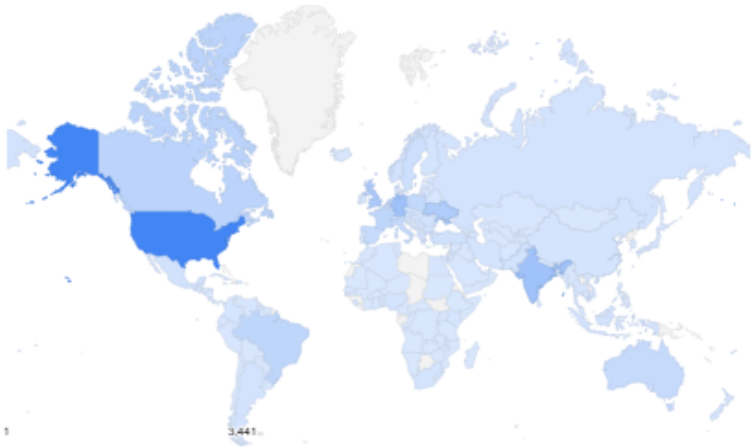


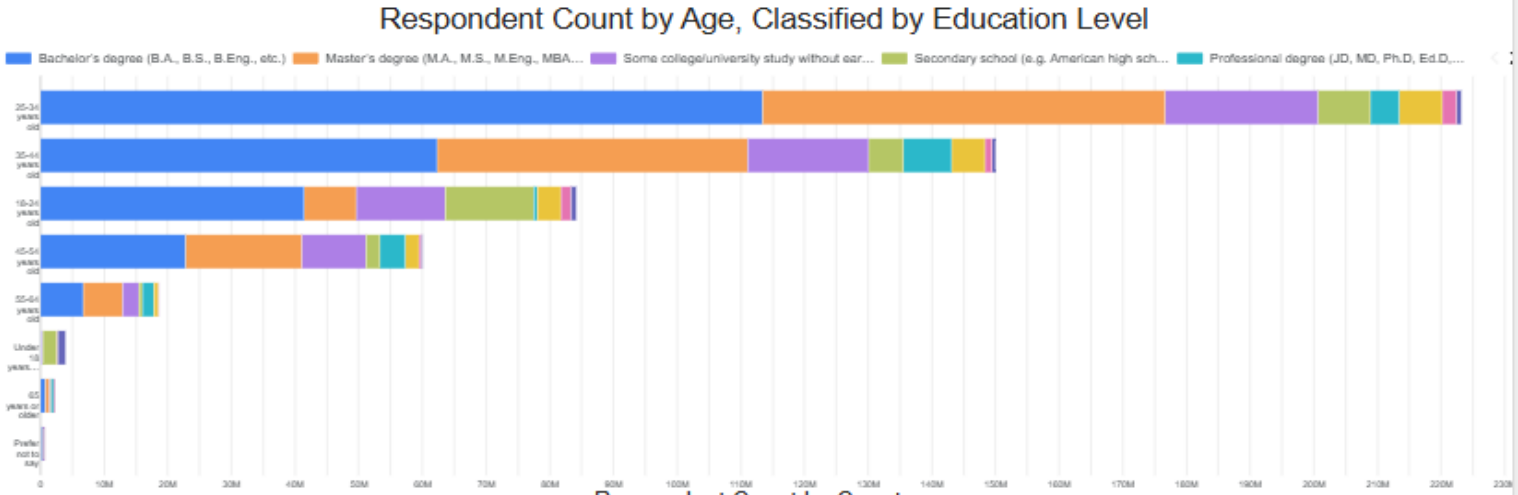


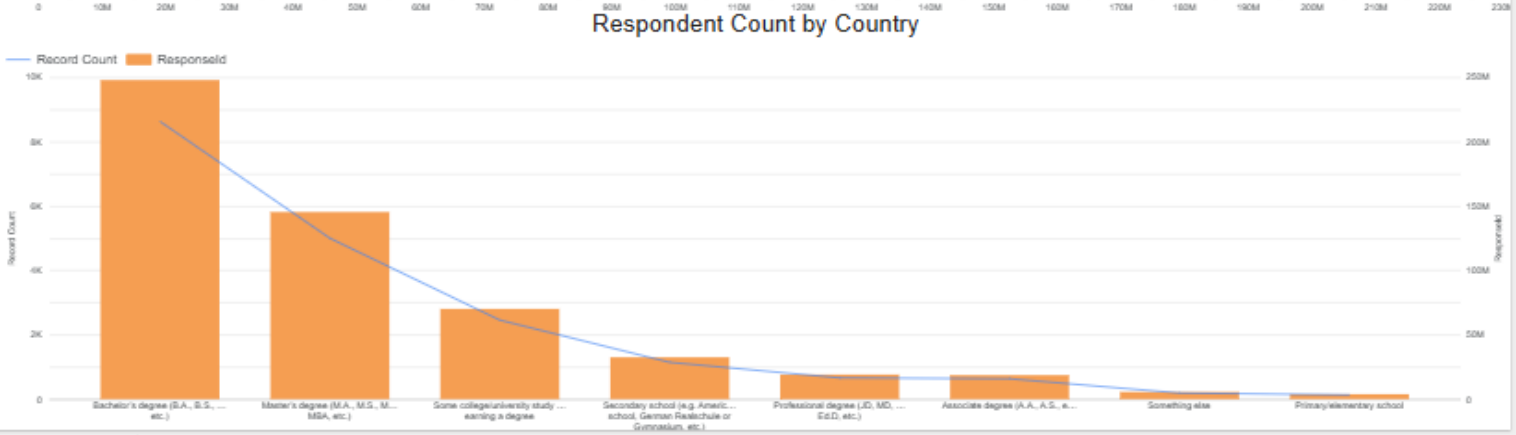
Slide 14

Shows the majority of devs are 25-34 years old









Slide 15: Dashboard Insights Summary

- Integrating the dashboards reveals that while the US leads in developer volume, emerging economies are adopting Cloud tools (AWS/Azure) at a faster rate. The younger demographic is also more likely to experiment with "Desired" technologies like Rust.

Slide 16 & 17: Overall Findings & Conclusion

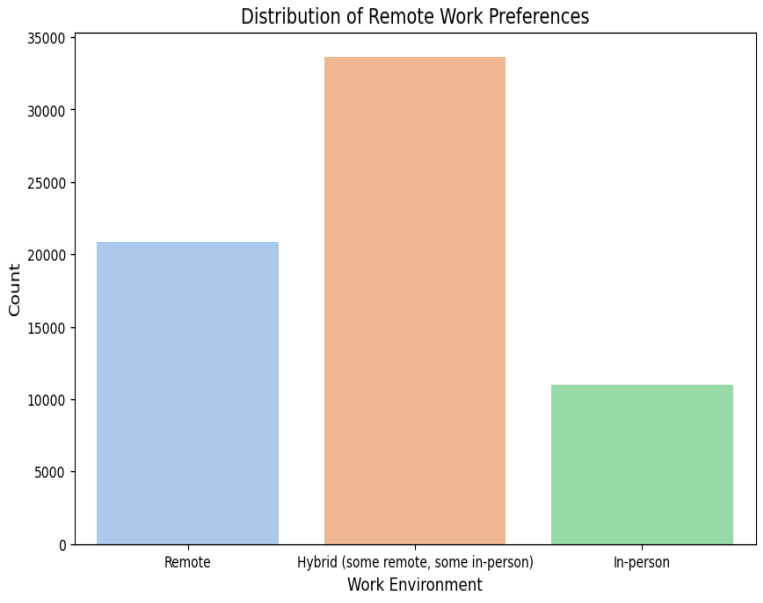
- **Conclusion:** The industry is in a "Hybrid" state; stable technologies (JS/SQL) provide the jobs, but emerging ones (Rust/AI) provide the growth.

Slide 16 & 17: Overall Findings & Conclusion

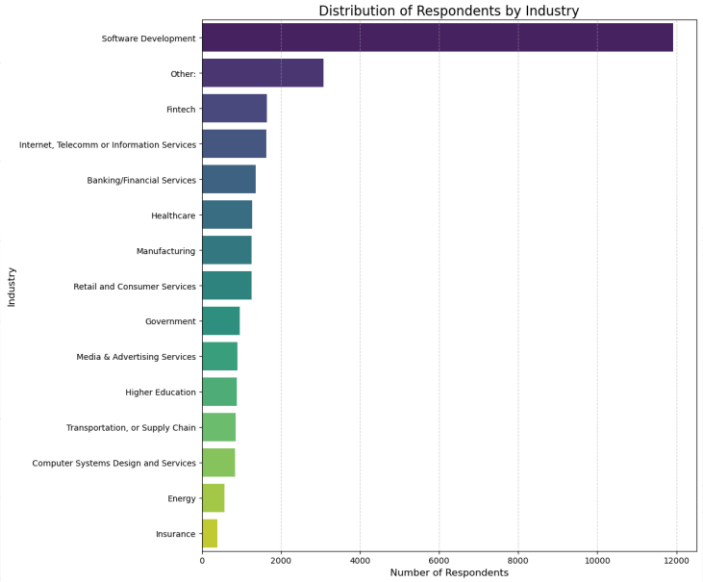
- **Key Recommendations:**
  1. Master **PostgreSQL** as it is the undisputed industry standard.
  2. Integrate **AI Tools** into the development workflow to stay relevant.
  3. Adopt a **Hybrid/Remote** mindset to attract top global talent.

Slide 18: Appendix (Additional Insights)

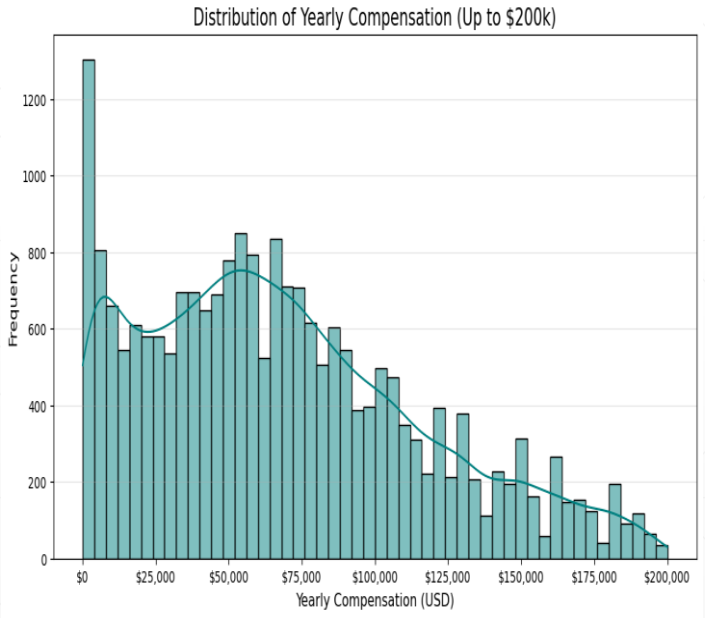
- Distribution of Remote Work Preferences



- Industry Distribution



- Distribution of Yearly Compensation



Slide 18: Appendix (Additional Insights)

- **Remote Work Preferences:** Most developers prefer **Hybrid** or **Remote** dne eht gnilangis , .are "ylno-eciffO" tcirts eht fo
- **Industry Distribution:** The **Software Development** and **FinTech** sectors are the largest employers, offering the highest compensation packages.
- **Salary vs. Experience:** Median salary peaks for those with 20-15years of experience, emphasizing that "Continuous Learning" is vital to maintaining salary growth.